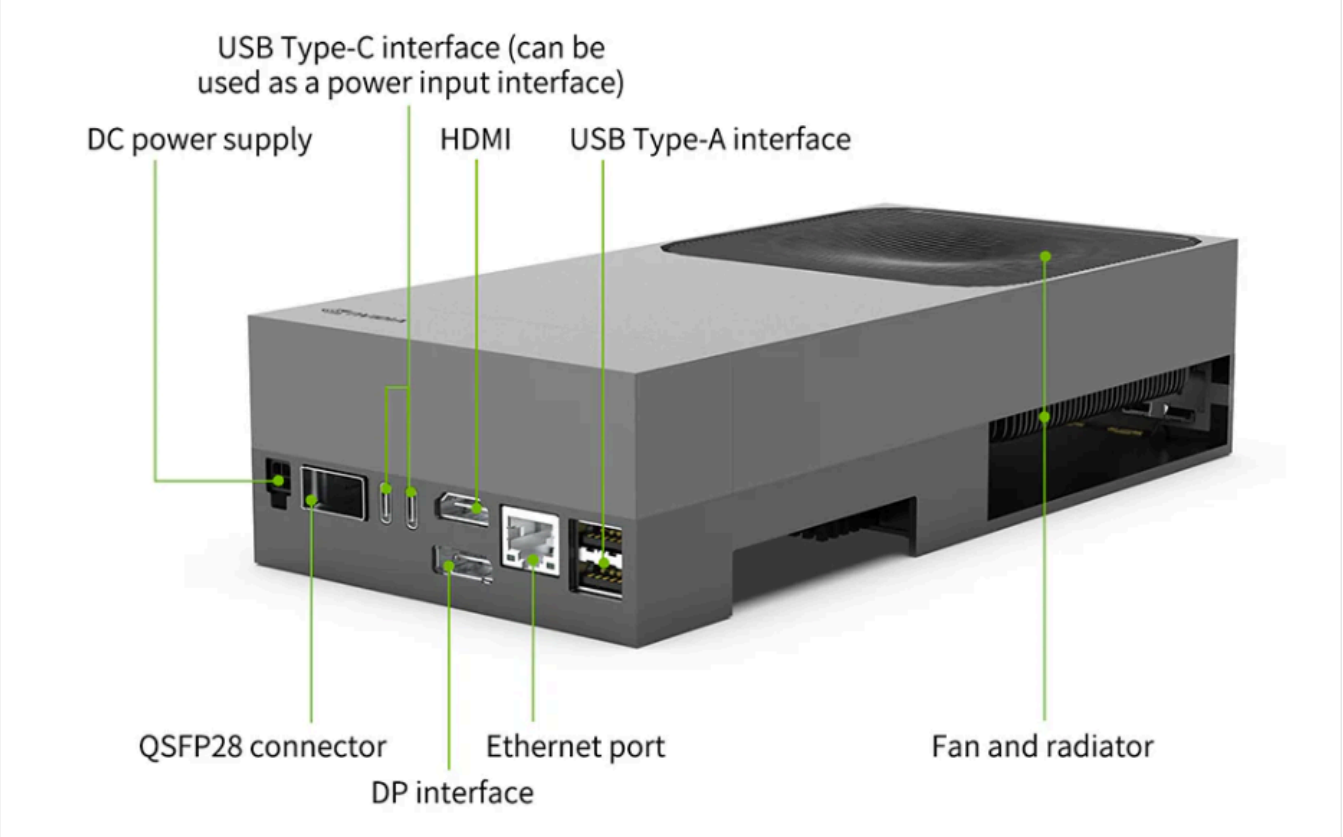
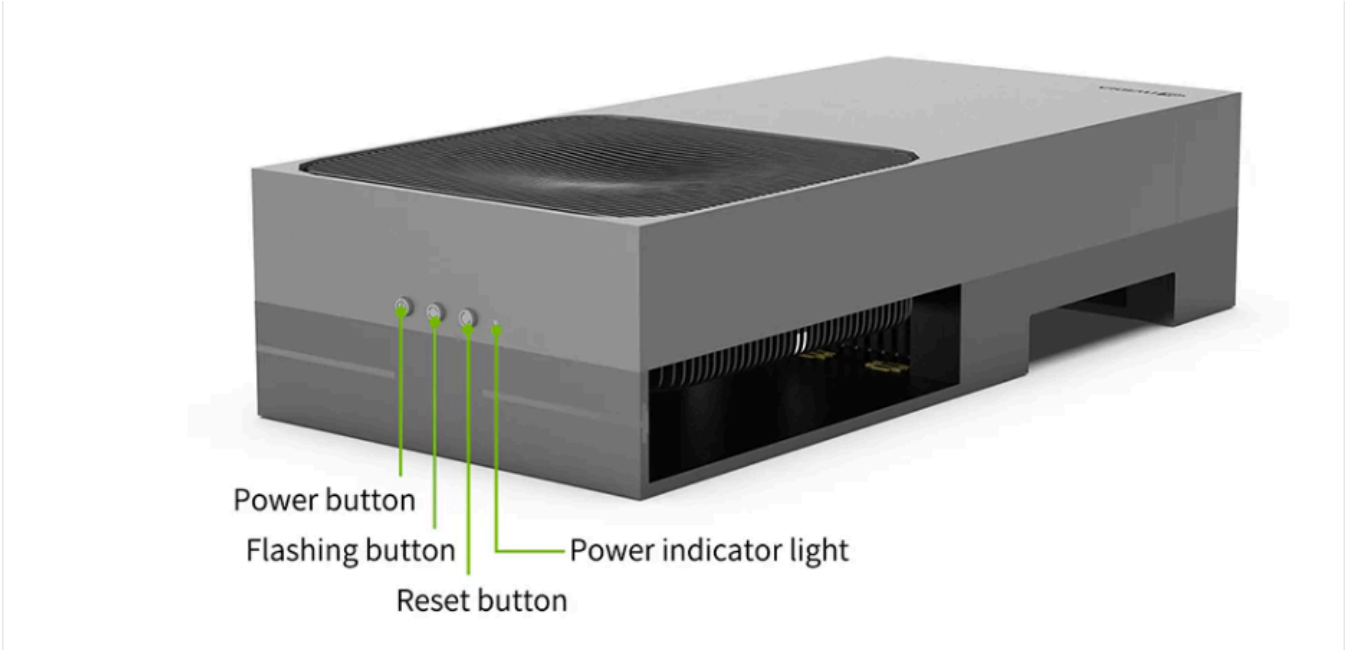
Jetson board introduction

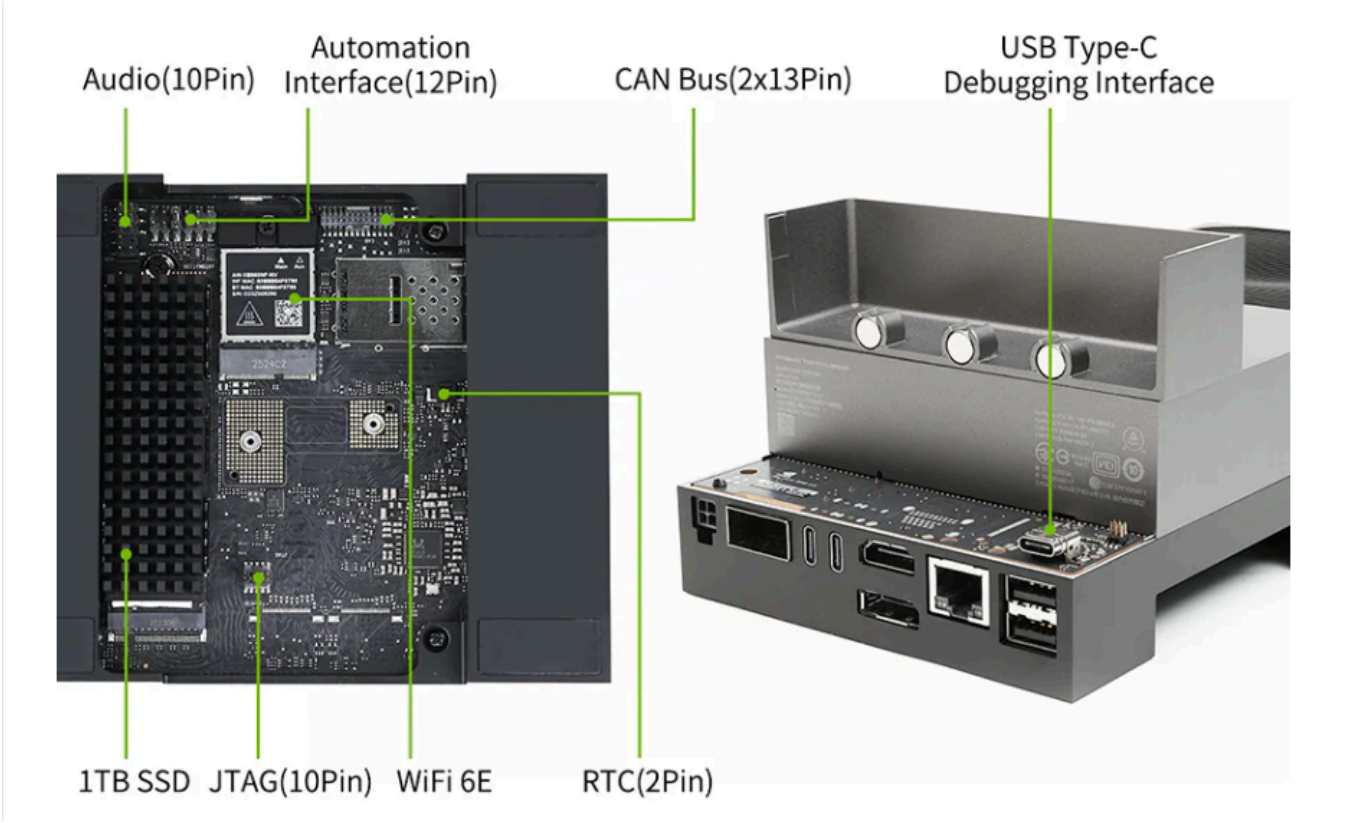
Jetson board introduction

1. Jetson AGX Thor Development Kit
2. Jetson AGX Thor module parameters
3. Motherboard Power Supply

1. Jetson AGX Thor Development Kit

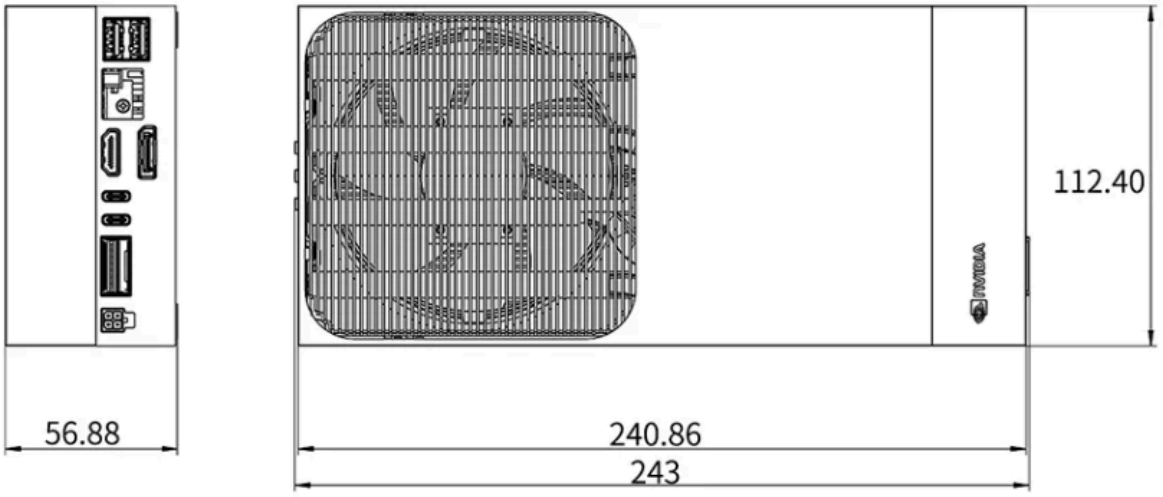



Power button
Flashing button
Reset button
Power indicator light


Audio(10Pin)
Automation Interface(12Pin)
CAN Bus(2x13Pin)
USB Type-C Debugging Interface
1TB SSD
JTAG(10Pin)
WiFi 6E
RTC(2Pin)

Dimensions

Units：mm



2. Jetson AGX Thor module parameters

JETSON Full Range Comparison

Compare only based on the core module

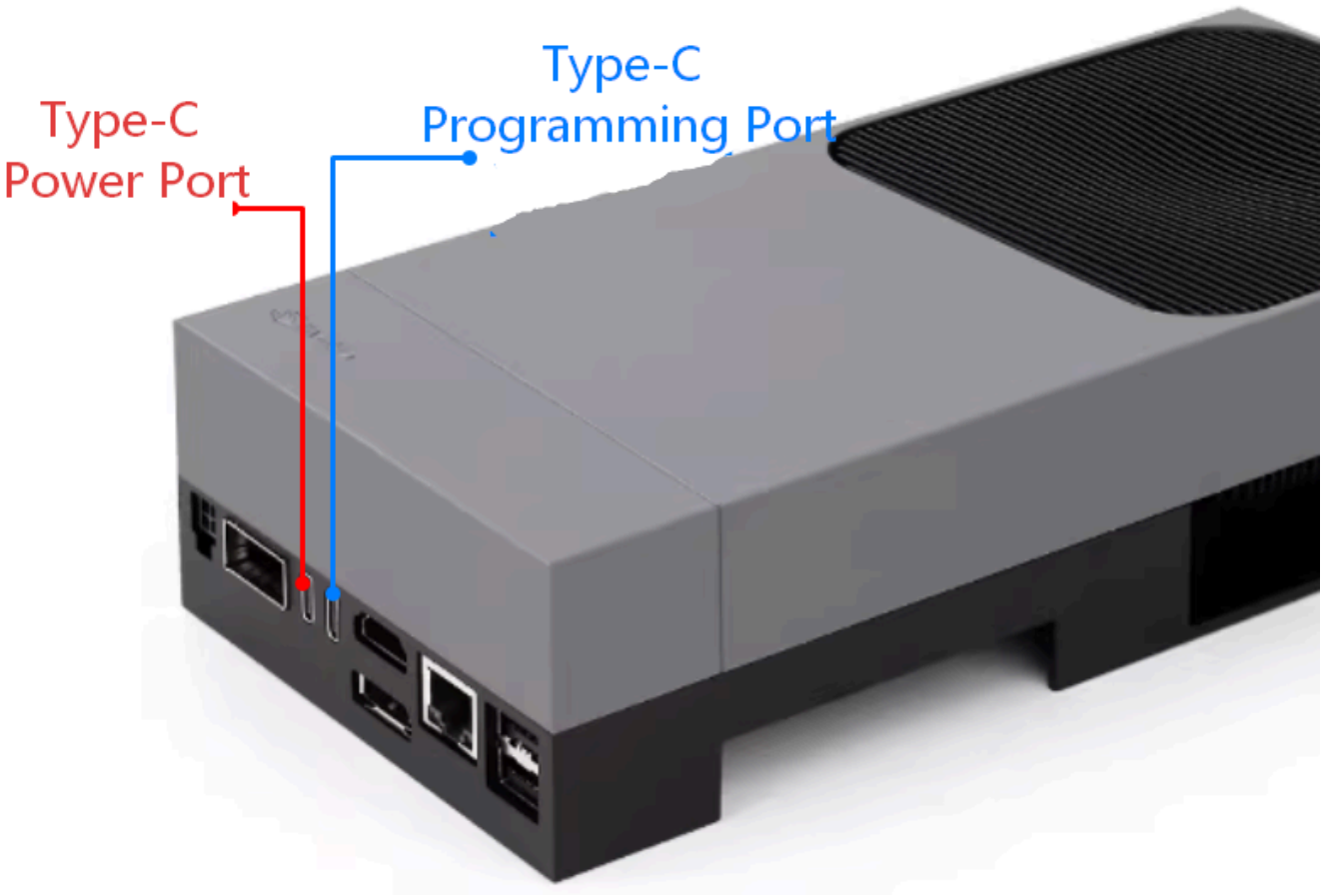
Model	Jetson Orin Nano Super 4GB	Jetson Orin Nano Super 8GB	Jetson Orin NX Super 8GB	Jetson Orin NX Super 16GB	Jetson AGX Orin 64GB	Jetson AGX Thor 128GB
AI Performance	34 TOPS (INT8)	67 TOPS (INT8)	117 TOPS (INT8)	157 TOPS (INT8)	275 TOPS (INT8)	2070 TFLOPS (FP4 —sparse)
GPU	512 Core NVIDIA Ampere architecture GPU with 16 Tensor Cores	1024 Core NVIDIA Ampere architecture GPU with 32 Tensor Cores	1024 Core NVIDIA Ampere GPU with 32 Tensor Cores		2048 Core NVIDIA Ampere GPU with 64 Tensor Cores	2561-core NVIDIA Blackwell architecture GPU with 96 5th-Gen Tensor Cores; With 10 TPC Multi-Instance GPU
GPU Max Frequency	1020 MHz		1173 MHz		1.3 GHz	1.57 GHz
CPU	6 Core Arm® Cortex®-A78AE v8.2 64 bit CPU1.5MB L2 + 4MB L3		6 Core NVIDIA Arm® Cortex A78AE v8.2 64 bit CPU 1.5MB L2 + 4MB L3	8Core NVIDIA Arm® Cortex A78AE v8.2 64 bit CPU 2MB L2 + 4MB L3	12 Core NVIDIA Arm® Cortex A78AE v8.2 64 bit CPU 3MB L2 + 6MB L3	14 Core Arm® Neoverse®-V3AE 64 bit CPU 1 MB L2 cache per core 16 MB shared system L3 cache
CPU Max Frequency	1.7 GHz		2 GHz		2.2 GHz	2.6 GHz
Vision Accelerator	/		2x NVDLA v2			1x PVA v3
DL Accelerator	/		1x NVDLA v2.0	2x NVDLA v2.0		/
DLA Max Frequency	/		1.23 GHz		1.6 Ghz	/
RAM	4GB 64 bit LPDDR5 51 GB/s	8GB 128 bit LPDDR5 102 GB/s 8 GB 128 bit LPDDR5 102.4 GB/s		16 GB 128 bit LPDDR5 102.4 GB/s	64GB 256 bit LPDDR5 204.8 GB/s	128 GB 256 bit LPDDR5X 273 GB/s
Storage	Supports External NVMe				64GB eMMC 5.1	1 TB NVMe M.2 Key M slot
Video Encode	1080p30, supported by 1–2 CPU cores		1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)		2x 4K60 (H.265) 4x 4K30 (H.265) 8x 1080p60 (H.265) 16x 1080p30(H.265)	6x 4Kp60 (H.265) 12x 4Kp30 (H.265) 24x 1080p60 (H.265) 50x 1080p30 (H.265) 48x 1080p30 (H.264) 6x 4Kp60 (H.264)

Video Decode	1x 4K60 (H.265) 2x 4K30 (H.265) 5x 1080p60 (H.265) 11x 1080p30 (H.265)	1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)	1x 8K30 (H.265) 3x 4K60 (H.265) 7x 4K30 (H.265) 11x 1080p60 (H.265) 22x 1080p30 (H.265)	4x 8Kp30 (H.265) 10x 4Kp60 (H.265) 22x 4Kp30 (H.265) 46x 1080p60 (H.265) 92x 1080p30 (H.265) 82x 1080p30 (H.264) 4x 4Kp60 (H.264)

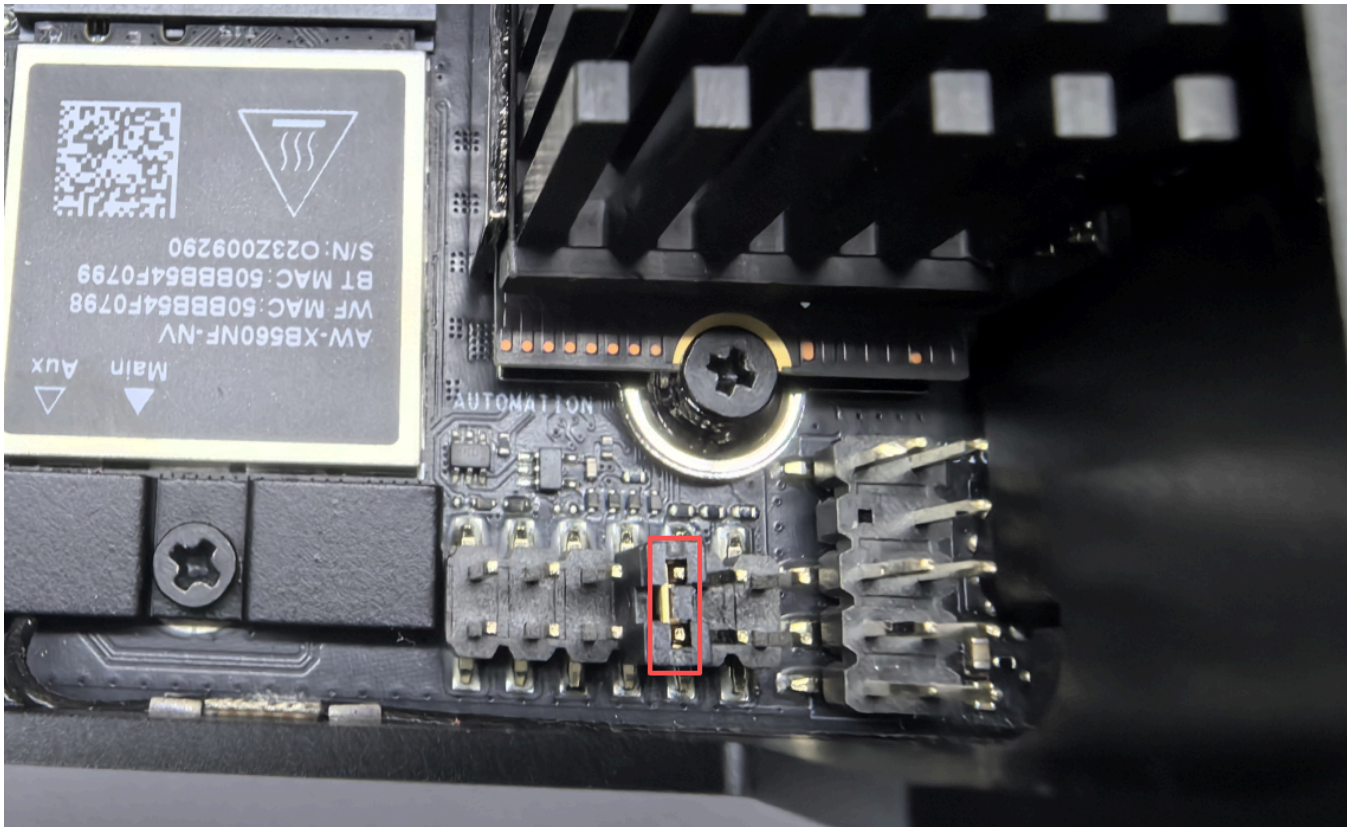
CSI Camera	Supports up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)				Up to 6 cameras (Supports 16 through virtual channels) 16 channel MIPI CSI-2 D-PHY 2.1 (up to 40 Gbps) | C-PHY 2.0 (up to 164 Gbps)	HSB camera via QSFP slot USB camera
PCIE*	1 x4 + 3 x1 (PCIe Gen 3, Root Port & Endpoint)				up to 2 x8 + 1 x4 + 2 x1 (PCIe 4.0、Root Port and Endpoint)	Support x4 PCIe 5.0 M.2 Key M Slots Support x1 PCIe 5.0 M.2 Key E Slots
USB*	3 X USB 3.2 Gen 2(10 Gbps) 3 X USB 2.0 interfaces				3x USB 3.2 2.0 (10 Gbps) 4x USB 2.0	2 X USB-A (3.2 Gen2) 2 X USB-C (3.1)
Networking	1x Gigabit Ethernet				1x GbE 1x 10GbE	1 X 5 GbE RJ45 interfaces 1 X QSFP28 (4 X 25 GbE)
Display	1x 4K30 Multi-mode DP 1.2 (+MST)/ eDP 1.4/HDMI 1.4**		1x 8K30 Multi-mode DP 1.4a (+MST)/ eDP 1.4a/HDMI 2.1		1x 8K60 Multi-mode DP 1.4a (+MST)/eDP 1.4a/HDMI 2.1	1 X HDMI 2.0b 1 X DisplayPort 1.4a
Other I/O	3x UART、2x SPI、2x I2S、4x I2C、1x CAN、DMIC 和 DSPK、PWM、GPIO				4x UART、3x SPI、4x I2S、8x I2C、2x CAN、PWM、DMIC and DSPK、GPIO	QSFP connectors M.2 Key E expansion Slots (WLAN/ BT, x1 PCIe, USB2.0, UART, I2C, I2S) M.2 Key M Connector (for NVMe Storage) PCIe x 4 channels、I2C、PCIe x 2 channels 2x 13 pin CAN Headers 2x 6 pin Automated headers LED JTAG connectors (2 x 5 Pin headers) 1x Fan headers – 12V、PWM and Tach Audio panel header (2 x 5 pin) Microfit Power socket RTC 2-Pin backup battery connector
Power	7W - 10W-25W	7W - 15W-25W	10W - 15W-25W-40W	10W - 15W-25W-40W	15W - 60W	40 W–130 W
Dimensions (Unit: mm)	69.6 x 45 260 pin SO-DIMM Connector				100 x 87 mm, 699-pin Molex Mirror Mezz Connector, Integrated heat spreader	243.19 x 112.40 x 56.88 (Thermal Plate (TTP) and optional fan or heatsink)

3. Motherboard Power Supply

Jetson AGX The official Thor kit includes a Type-C power adapter (maximum 240W). Both Type-C ports on the motherboard can be used as power inputs, but only the one closer to the HDMI port (blue text) can be used for system flashing. Therefore, generally, the Type-C port closer to the QSFP28 port (red text) is used for power, and the Type-C port closer to the HDMI port (blue text) is used for flashing.



By default, the AUTOMATION header located at the bottom of the motherboard is configured as shown in the image below; with this configuration, the system will automatically power on and boot into the OS immediately upon being plugged in. If this jumper cap is removed, the system will not power on automatically when connected to power; instead, you will need to press the power button manually to turn it on.



The Jetson AGX Thor motherboard requires a connected display to properly render the system desktop. For first-time use, it is recommended to connect an external monitor, mouse, and keyboard.